Algebra 1 | Name____________________
Unit 5 | Date____________________
Lesson 2 Homework | Period____________________

1. Solve the equation and verify the solution using substitution.

$$-\frac{1}{8}y + 7 = 39 - \frac{3}{4}(-8 - 3y)$$

The formula for calculating the slope of a line from the points (x_1, y_1) and (x, y) is given by the following equation for questions 2-3, $m = \frac{y-y_1}{x-x_1}$.
The point-slope form of a line is as follows: $y - y_1 = m(x - x_1)$

2. Which operation (addition, subtraction, multiplication, or division) can be used to rearrange the slope formula into the point-slope form of a line?

3. Use your answer from question 2 to rearrange the slope formula to get the point-slope form of a line.

Use the standard form of a line is given by the formula $Ax + By = C$ for questions 4 – 5.

4. Rearrange this formula to solve for the variable, y.

5. Based on your answer to question 4, what is the slope of the line you found? *How do you know?* Explain your thinking in words.

For questions 6 – 7, rearrange the formulas to isolate the specified variables.

6. The formula $K = \frac{1}{2}mv^2$ is used to find the kinetic energy of an object in motion using its mass, m, and its velocity (speed), v. Solve the equation for the mass, m.

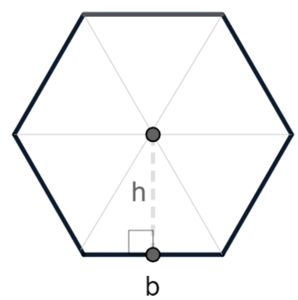
7. The formula for the volume of a cylinder is given by $V = \pi r^2 h$, where r is the radius of the base of the cylinder and h is the height. Solve the equation for the height, h.

8. The formula for the area of a regular hexagon, A, is $A = \frac{3\sqrt{3}}{2} \cdot s^2$, where s is the side-length.

Lashonda rearranged the area formula to isolate the side-length, s. She knows that the area of the hexagon is 64.95. Her work is shown below. Complete the table to explain each step of Lashonda's work.

Lashonda's work:	**Explanation of the steps**
$A = \frac{3\sqrt{3}}{2} \cdot s^2$	Formula for the area of a regular hexagon
$2 \cdot A = \cancel{2} \cdot \frac{3\sqrt{3}}{\cancel{2}} \cdot s^2$ $2A = 3\sqrt{3} \cdot s^2$	Multiply both sides of the equation by 2
$\frac{2A}{3\sqrt{3}} = \frac{\cancel{3\sqrt{3}} \cdot s^2}{\cancel{3\sqrt{3}}}$ $\frac{2A}{3\sqrt{3}} = s^2$	
$\sqrt{\frac{2A}{3\sqrt{3}}} = \sqrt{s^2}$ $\sqrt{\frac{2A}{3\sqrt{3}}} = s$	
$\sqrt{\frac{2(64.95)}{3\sqrt{3}}} = s$	
$5 \approx s$	

Xavier is Lashonda's classmate. He said that he did not need the formula for the area of a hexagon to find the side length. He connected the vertices of the hexagon to form 6 equilateral triangles, shown in the figure to the right. He used the formula for the area of a triangle, $A_{\text{triangle}} = \frac{1}{2}bh$, to create the following formula for the area of a regular hexagon: $A_{\text{hexagon}} = 6\left(\frac{1}{2}bh\right)$.
Xavier got stuck because he did not know the height, h, of the triangles!



9. Rearrange Xavier's formula, $A_{\text{hexagon}} = 6\left(\frac{1}{2}bh\right)$, to solve for the height, h.

10. Substitute the area, 64.95, and the side-length Lashonda found, 5, to find the height of Xavier's triangles.